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Upper Cretaceous paleomagnetic data from Shikotan Island, Kuril Arc: Implications for plate kinematics

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Abstract

Maastrichtian tuffaceous sandstones and siltstones were sampled from four sites in the northern part of Shikotan Island in the Lesser Kuril Islands. A characteristic component (ChRM) isolated from most samples passes the reversal and fold tests and is most probably a primary remanence. The ChRM mean direction ($D = 334°$, $I = 56°$, $k = 36$, $\alpha_{95} = 3.2°$) is $14° \pm 3.5°$ steeper than the Pacific reference direction and $13 \pm 4°$ more gentle than the Eurasian reference direction. We assume that the Lesser Kuril Islands together with their southwestern extension into the Nemuro Peninsula, Hokkaido were originally an island arc situated at about 36°N in the central-west Pacific. Soon afterwards, probably by the end of the Late Cretaceous, the Nemuro–Shikotan island arc became inactive and started moving with the Pacific plate. In the Miocene, about 15 m.y. ago, the Nemuro–Shikotan island arc collided with the Eurasian plate, overrode the subduction zone, and occupied its present-day position on the Pacific side of the Kuril island arc. Although this scenario fits the available paleomagnetic data and kinematics of the Pacific plate itself, it disagrees somewhat with the kinematic models for the North Pacific as a whole. In particular, by the end of the Late Cretaceous, the northern margin of the Pacific plate together with the extinct Nemuro–Shikotan island arc should have been in an area close to the transform or ridge-type Kula–Pacific boundary. The available paleomagnetic data from Northeast Asia, however, indicate that active islands arcs existed in mid-northern latitudes of the modern Pacific Ocean in the Cretaceous.

1. Introduction

It is now widely recognized that continental masses surrounding the North Pacific are an agglomerate of terranes transported from far away and later welded into adjacent plates. Paleomagnetic data are paramount in deciphering the evolution of this region. The magnitudes of horizontal movements, though still far from being clear, are much better known for the North America margin where numerous paleomagnetic data are available [e.g., 1–3]. The situation is much worse for the Northwest Pacific periphery, where paleomagnetic results have only just started to appear [4–7]. Those already available from the Northwest Pacific point to large-scale displacements of some terranes and a rather complex evolution of Northeast Asia.

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Below are presented new paleomagnetic data on Upper Cretaceous volcanosedimentary rocks from the southern part of the Kuril island arc which we hope may place additional constrants on the tectoncs of the Northwest Pacific.

2. Geological setting and sampling

The Lesser Kuril Islands and the the Nemuro Peninsula in the easternmost part of Hokkaido are an area without present-day volcanic activity on the Pacific side of the Kuril island arc (Fig. 1). Upper Cretaceous volcanics, however, are common in this area, and this is considered to represent a remnant of a Late Cretaceous island arc which had originated on the oceanic crust, as is evident from the pyroxenite, gabbro and serpentinite xenoliths in Cretaceous basalts [8].

Upper Cretaceous rocks are found in two complexes, the upper complex being thrust on top of the lower one [9,10]. Basalts, tuffaceous conglomerates and ~800 m thick sandstones bearing *Inoceramus* of Campanian age (Matakatan Formation) are exposed at the base of the lower complex. These rocks are conformably overlain by unmetamorphozed tuffaceous siltstones and sandstones of the Malokuril Formation, which is ~400 m thick and bears the Maastrichtian species of *Inoceramus* [11]. The upper thrust sheet consists mainly of ophiolites, and is overlain by Maastrichtian–Paleocene conglomerates [12]. On the Nemuro Peninsula, the Notsukamappu and Nemuro Formations are counterparts of the Matakotan and Malokuril Formations respectively. Monzonite and trachydolerite sills present among the Notsukamappu and Nemuro Formations were dated at 84–88 Ma by the K-Ar method [13]. Debris from these volcanics are found in the overlying Danian Choboshi Formation. A paleomagnetic study of these sills yielded a mean direction of stable magnetization of $D = 348°$, $I = 54°$ ($k = 22$, $\alpha_{95} = 5.9°$) [13]. Also reported, but unfortunately only as an abstract, is a paleomagnetic result ($D = 330°$, $I = 53°$, $k = 31$, $\alpha_{95} = 6°$) on pillow lavas of the Matakatan Formation from Shikotan Island [14].

Table 1
Paleomagnetic results for the Maastrichtian Malokuril Formation

Site	N	$A°/d°$	*In-situ*				Tilt-corrected			
			$D°$	$I°$	k	$\alpha_{95}°$	$D°$	$I°$	k	$\alpha_{95}°$
1-a	5/4	90/10	333	52	28	13.4	348	56	31	12.6
1-b	11/11	90/40	298	28	24	8.6	324	59	28	8.0
1-c	13/13	100/25	313	37	49	5.6	333	56	50	5.5
2	10/9	228/13	345	46	32	8.3	330	51	32	8.3
3	17/14	162/16	340	38	33	6.5	339	54	40	5.9
4	6/2	230/12	355	54	–	–	338	59	–	–
Overall means										
53 samples			324	40	15	5.0	334	56	36	3.2
6 mean vectors			328	44	19	13.2	335	56	216	3.9
Tests										
Reversal test (site 3)										
$F_{(2,26)} = 3.4$							$f = 1.5$			
Fold test										
$F^{*}_{(10,94)} = 1.9$			$f = 13.5$				$f = 1.2$			
$F^{**}_{(8,92)} = 2.1$			$f = 15.5$				$f = 1.5$			

* The test is performed for all six means; ** the worst-defined mean for site 4 is omitted. Site = number of sites (numbers) and groups of bedding poles (letters, for site 1); N = number of samples studied/accepted; A = azimuth of dip; d = angle of dip; D = declination; I = inclination (all the data are inverted to normal polarity); k = concentration parameter; α_{95} = 95% circle of confidence; F = 95% critical value of F statistic with the numbers in parentheses denoting degrees of freedom; f = calculated values of F statistic. The underlined result was used for interpretation.

In general, the Upper Cretaceous lower complex of Shikotan gently dips to the southeast, but some folds are locally present. On Shikotan, the deformation age is poorly constrained by an angular unconformity between the Upper Cretaceous–Paleocene and Quaternary rocks. There is, however, an observation reported of gabbro blocks, presumably from the upper thrust sheet, and synsedimentary folding in the sedimentary rocks at the top of the Malokuril Formation. These observations have been interpreted as the result of thrusting which took place in the Maastrichtian or slightly later [9,10]. We tentatively hypothesize that the larger folds observed in the Malokuril Formation may be of the same age. The assumption that deformation occurred long before the Quaternary is corroborated by angular unconformities between the Oligocene and Up-

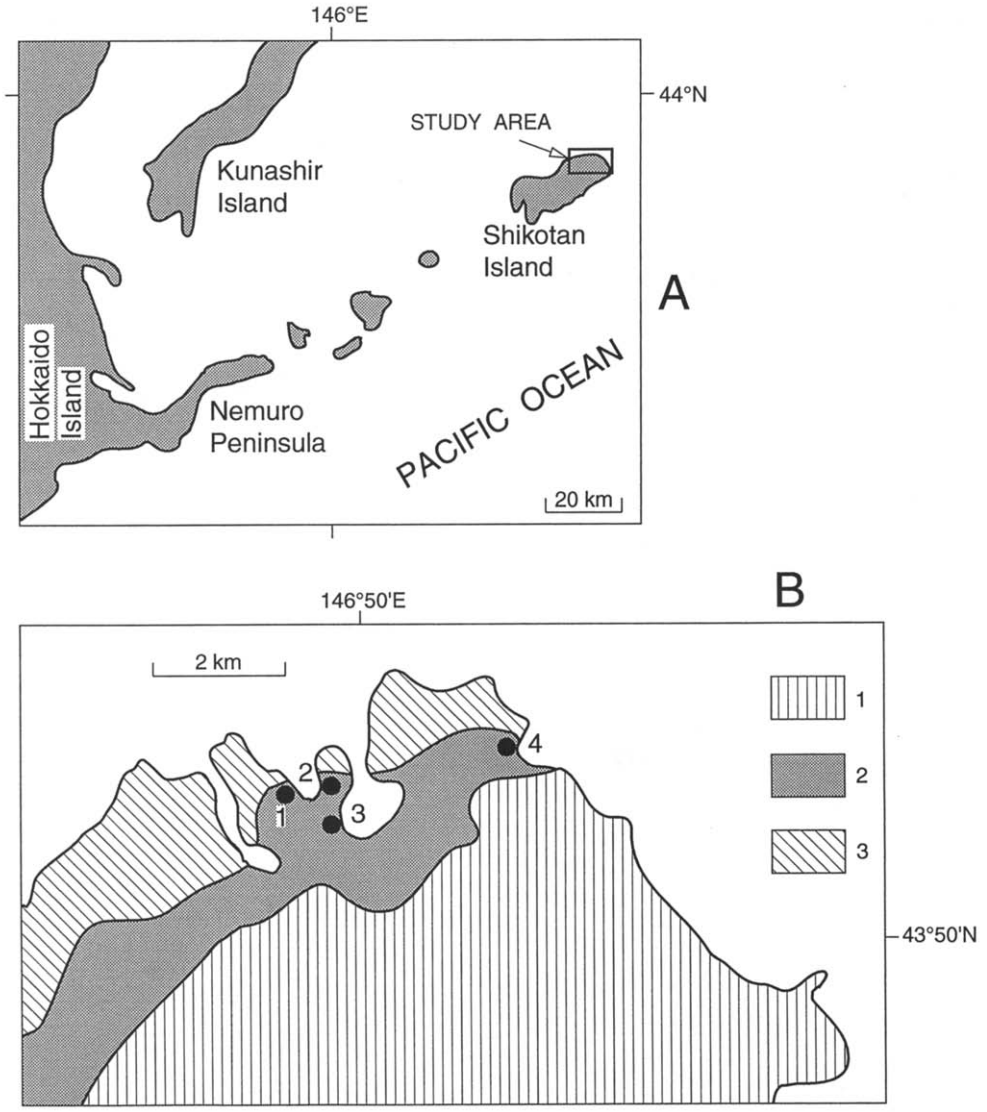
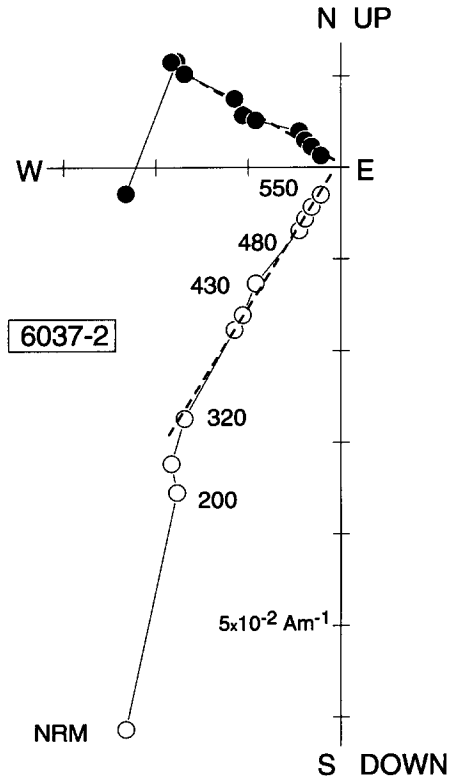
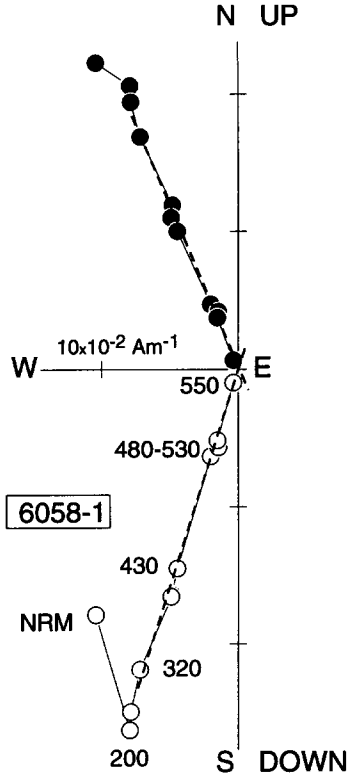
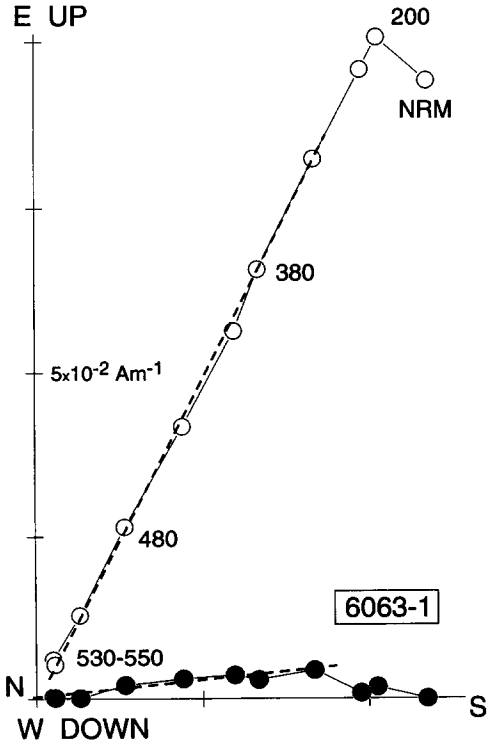
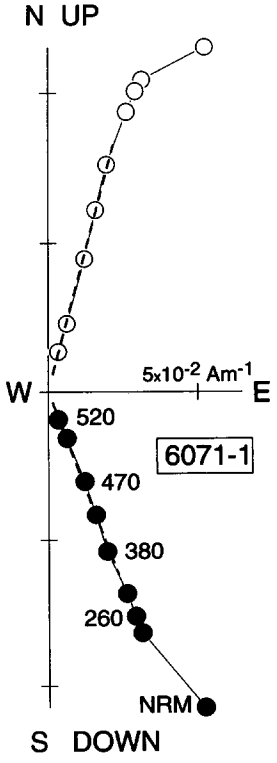


Fig. 1. (A) Study area and (B) schematic geological map of the northern part of Shikotan (simplified after [10]) with sampled sections shown as •. *1* and *2* = lower complex: *1* = Matakotan Formation, *2* = Maloruril Formation; *3* = upper complex.


N UP
W
E
550
480
430
6037-2
320
200
5x10-2 Am-1
NRM
S DOWN


N UP
10x10-2 Am-1
W
E
550
480-530
6058-1
430
NRM
320
200
S DOWN


E UP
200
NRM
380
5x10-2 Am-1
480
6063-1
530-550
N
S
W DOWN


N UP
5x10-2 Am-1
W
E
520
6071-1
470
380
260
NRM
S DOWN

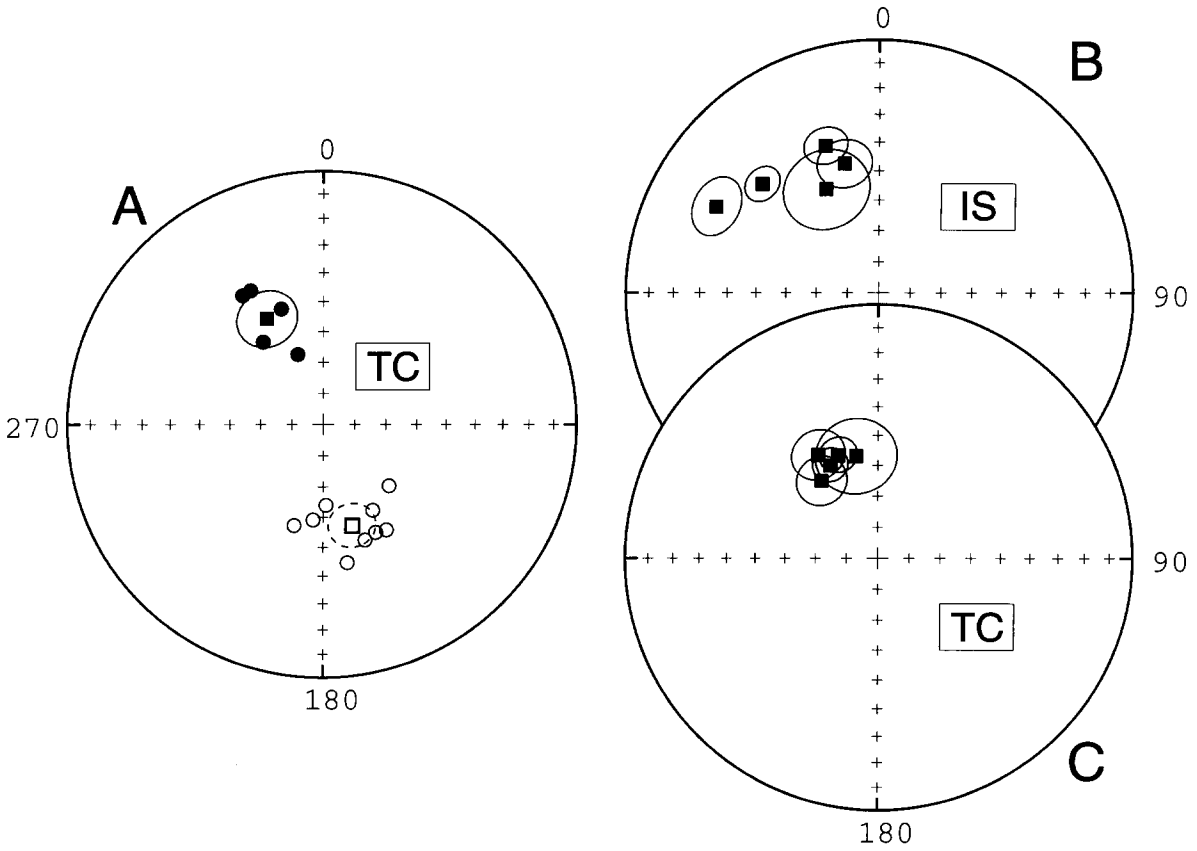


Fig. 3. Equal-area projection of (A) tilt-corrected ChRM directions from site 3, and (B) *in-situ* and (C) tilt-corrected mean directions from the Malokuril Formation. ○ and ● = sample directions; □ and ■ = mean vectors together with their confidence circles. Solid symbols and solid lines = projections onto lower hemisphere; open symbols and dashed lines = projections onto upper hemisphere.

per Cretaceous and between the Miocene and Paleocene rocks [11] described on the Nemuro Peninsula in the eastern part of Hokkaido on the southward extension of the Shikotan structures.

In 1981, paleomagnetic samples were collected from fine-grained tuffaceous rocks of the Malokuril Formation on northern Shikotan (Fig. 1B). Sites 1–3 are from two limbs of the SW–NE trending syncline with dips from 10° to 50°. One hand sample oriented with the magnetic compass was taken from each layer sampled. The true thickness at each site was up to 70 m, and the total thickness studied certainly exceeds 100 m because different levels of the formation were sampled.

3. Methods and results

Two specimens from each sample were subjected to stepwise thermal demagnetization in home-made ovens with a residual field of about 10 nT. All measurements were made on a JR-4 Czechoslovakian spin magnetometer. The fold test, based on the comparison of mean vectors, was carried out in the following way: All normals

Fig. 2. Representative orthogonal plot (tilt-corrected) data from the Malokuril Formation. Dashed lines show the adjustment of linear segments; ● = projections onto horizontal plane; ○ = projections onto vertical plane; temperatures are in °C.

to bedding planes were drawn on the stereonet and then their distribution was divided into a number of approximately isometric non-overlapping groups. Three such groups were recognized at site 1 (a,b,c in Table 1) while the other three sites were treated as monoclinal sections. The corresponding mean vectors were compared with the aid of the statistical F criteria both before and after tilt correction [15].

The whole collection was heated in five steps up to 400°C. About 60% of the collection was completely stepwise demagnetized. After analyzing the demagnetization data we found that the samples from sites 1–3 responded quite well to thermal cleaning and, apart from a low-temperature scattered remanence, the NRM could be accounted for by a single component showing rectilinear decay to the origin above 300° (Fig. 2). The directions of this characteristic component (ChRM) are in perfect agreement for two specimens from a given sample. The results after (i) complete demagnetization and component analysis [16] and (ii) after heating to 400° are very similar. For instance, fourteen completely demagnetized samples from site 1 have mean directions of $D = 338°$, $I = 57°$ (K = 26, $\alpha_{95} = 7.4°$) and $D = 342°$, $I = 54°$ ($k = 32$, $\alpha_{95} = 6.5°$) respectively. Seeing as complete demagnetization and blanket cleaning at 400° yielded very similar results, both datasets were pooled. At site 4, however, the rocks proved to be heavily overprinted by a remanence with an *in-situ* direction close to the present-day field, and, although all the samples were completely demagnetized, the ChRM was isolated from only two samples. The ChRM has a normal polarity at sites 1, 2 and 4, whereas both polarities were found at site 3. Mean directions of the two polarities are antipodal, as is evident from the reversal test (Fig. 3 and Table 1). The reversed directions were inverted through the origin and the fold test [15] was applied to six mean directions. The site means (sites 2–4) and group means from site 1 certainly pass the fold test (Table 1). The mean for site 4 is based only on two unit vectors, but its omission does not change the conclusion (Table 1).

Thus, a unicomponent bipolar prefolding magnetization is isolated from unmetamorphozed tuffaceous rocks of the Late Cretaceous Malokuril Formation. The rather large thickness studied and the presence of both polarities imply that a long time span is covered by the collection, and it is likely that all types of paleomagnetic noise have been properly averaged out. Taking into account the possibility that the folding and thrusting on Shikotan are contemporaneous (i.e., Late Cretaceous in age), the close similarity between Shikotan and the Nemuro Peninsula and the presence of pre-Oligocene deformation on the latter, the ChRM is likely to be primary, especially when the reversal test is taken into account.

4. Interpretation and discussion

The overall mean direction from the Malokuril Formation was compared to the Eurasian, North American and Pacific reference data. Two rather different versions of the Cretaceous part of the Eurasian polar wander path [17,18] were used, but the Eurasian reference directions proved to be fairly similar for Shikotan, while the Pacific reference direction [19] is quite different. The kinematic parameters, namely flattening F and rotation R, together with the error limits [20], are presented in Table 2. Seeing as the confidence area around the Pacific mean pole is an oval, we decided to replace this oval with a circle with a radius equal to the major semi-axis, and to use

Table 2
Kinematic parameters

Pole	$I°$	$F° \pm dF°$	Pl	$D°$	$R° \pm dR°$
1. Eurasia	74	18 ± 3	60	4	-30 ± 8
2. Eurasia	69	13 ± 4	52	14	-40 ± 8
3. N.America	70	16 ± 4	54	17	-43 ± 8
4. Pacific	45	-14 ± 4	27	348	-14 ± 6

Pole = mean paleomagnetic poles used for computation of reference direction: (1) = Eurasian Cretaceous mean pole [17]; (2) = 70 Ma Eurasian pole [18]; (3) = North American Late Cretaceous pole [18]; (4) = Maastrichtian Pacific pole [19]. D and I = declination and inclination of the reference direction; F = flattening; dF = 95% confidence semi-interval; R = rotation; dR = 95% confidence semi-interval.

this value, $A_{95} = 3.0°$. A considerable part of Northeastern Eurasia is often thought to have belonged to the North American plate (NAP) in the Late Cretaceous; the plate boundary is placed within the Mesozoic and Cenozoic foldbelts between Siberia in the west and the Bering Strait in the east. The Late Cretaceous NAP reference direction recalculated from the 70 Ma mean pole [17] is fairly similar to the Eurasian direction and the latter was therefore used subsequently.

The large difference between the Eurasian and Pacific reference inclinations reflects the convergence of these two plates (more precisely, a part of the total convergence which is reflected in paleomagnetic data). The Shikotan paleomagnetic inclination does not agree with that expected for either of the reference fields (Fig. 4 and Table 2). It is significantly lower than those expected for the Eurasian reference direction; the values of flattening F, however, depend somewhat on the pole used. In contrast, the measured inclination is significantly steeper than that expected on Shikotan for the Pacific reference pole (Fig. 4 and Table 2). This pattern infers a movement of the Nemuro–Shikotan terrane (NST) with respect to the main plates. Very important is the fact that the excellent agreement between our result on sedimentary rocks and those on volcanics from the Nemuro Peninsula [13] and Shikotan [14] implies that both types of rocks can be regarded as faithful recorders of the ambient field.

Most kinematic schemes suggested so far assume that oceanic plates in the North Pacific have always subducted approximately northward under Eurasia and North America [21–23]. Thus, the original position of the NST on the Pacific plate is unacceptable because this terrane is an island arc. The magmatic activity, however, ceased soon after deposition of the studied rocks (the Malokuril Formation is the youngest on Shikotan, whereas the youngest volcanics on the Nemuro Peninsula are Danian in age [11]). We assume that the NST could have welded onto the Pacific

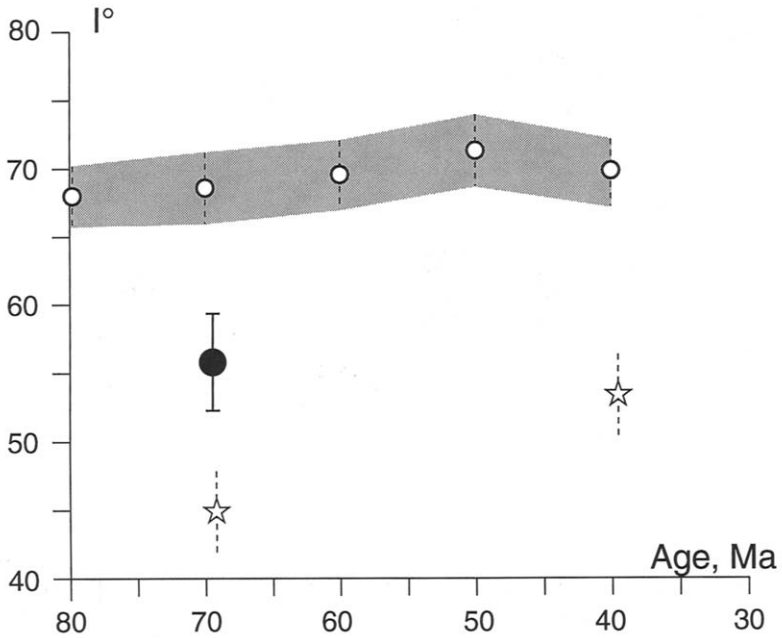


Fig. 4. Inclination versus age plot of paleomagnetic results reference data together with error bars (shown as vertical lines). ○ = our result from Shikotan; ☆ = Pacific reference inclinations [19]; ○ = Eurasian reference data [18]. For the latter, the confidence band is shaded.

plate by the very end of the Cretaceous and moved as an integral part of this plate since that time.

At present, the NST belongs to the Eurasian plate, and the collision of these two elements certainly occurred some time ago. By incremental removal of the Pacific trajectory backward in time, we find that that the observed latitude is compatible with the Pacific trajectory in a time window from 5 to about 22 Ma; the best fit of the observed latitude with that predicted by the Pacific kinematics is observed if the collision of the NST with Eurasia took place about 15 Ma ago (Fig. 5). This finding is in agreement with the geological data. For instance, studies on Hokkaido have shown that this island is an accretionary complex which resulted from the Miocene collision of several island arcs, including the NST [24,25]. Thus, the paleolatitude difference between Shikotan and the Pacific plate is accounted for by the continued subduction of this plate under Eurasia after the NST sutured to Eurasia in the Miocene (Fig. 4). In contrast, the paleolatitude difference between Shikotan and Eurasia is most logically explained by the subduction of the Pacific plate under Eurasia while the NST was a part of the former.

If the above model is correct we can restore the NST to its original position, or rather to a position from whence this terrane began moving as a part of the Pacific plate. This position is shown as a small white rectangle at about 36°N, 200°E in Fig. 5. The trajectory of the Pacific plate before 56 Ma, however, is considered to be less reliable than after 56 Ma [23]. Therefore, it is better to say that the original position of the NST lay at 36°N and at a longitude to the east of 180°E. The same point also marks the northern boundary of the Pacific plate at the end of the Cretaceous, because it is unlikely that an island arc could have been incorporated into the Pacific plate far from the boundary. Which plate was situated immediately to the north remains an open question.

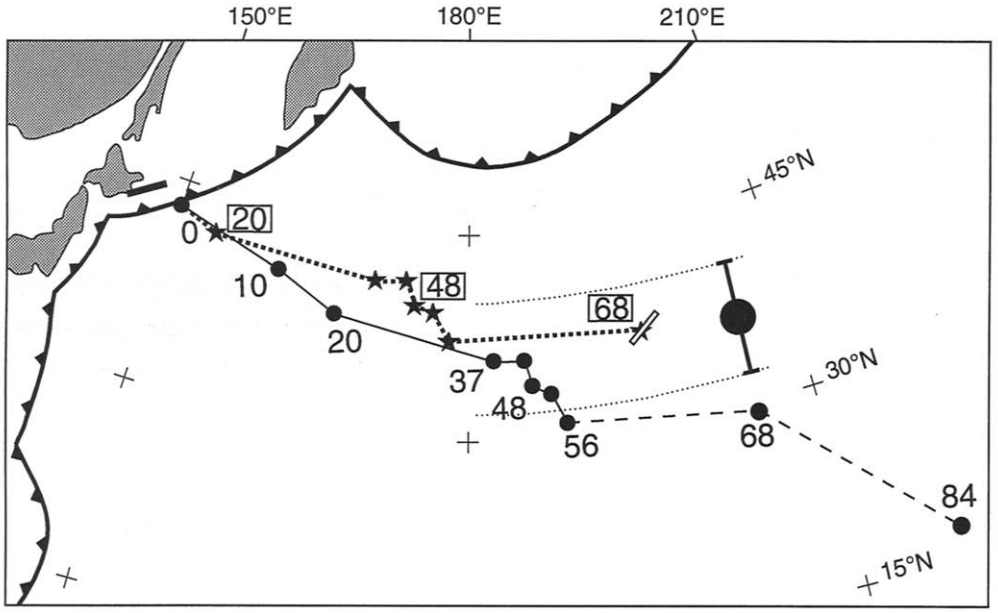


Fig. 5. A kinematic model of the Nemuro–Shikotan terrane. A trajectory (•) of the Pacific plate (Eurasia fixed) for the last 84 Ma [23] is shown; solid and dashed lines denote, respectively, reliable and less reliable parts of the trajectory [23]. ⋆ and heavy dotted line = the trajectory of the NST corrected for the collision of this terrane with Eurasia at about 15 Ma. The NST paleolatitude and its confidence band are shown as a large dot with error bar and light dotted lines respectively. The inferred original and present-day positions of the NST are shown as small white and black rectangles, respectively (see text).

However, our model, and the conclusion on the position of the northern margin of the Pacific plate in particular, somewhat contradicts other models of Pacific kinematics [21–23] which predict that this part of the globe was occupied by the Kula plate or was close to the transform or ridge-type Kula–Pacific boundary. Close juxtaposition of convergent and divergent plate boundaries does not seem to be likely. An assumption that the NST could have belonged to the Kula plate (or yet another plate) requires a complicated pattern of NST kinematics, including a change in the NST's 'host plate' on at least one occasion more than we have identified. Our scheme described above, therefore, appears much more simple than this complex scenario.

With respect to the Eurasian and North American plates, the NST data are rotated counterclockwise (Table 2). In principle, this rotation may be related to the opening of the Sea of Japan and the bending of the Japanese Islands in the Miocene. On the other hand, the mean declination from the Nemuro Peninsula ($D = 348°$) agrees with the Pacific reference direction, whereas the Shikotan is rotated $14° \pm 9°$ counterclockwise with respect to the Nemuro Peninsula. This small rotation is presumably related to internal deformation of the NST during its oblique collision with Eurasia [26]. Apart from this local movement, the NST appears to be unrotated with respect to the Pacific plate. This is in agreement with the model proposed above of the NST moving together with the Pacific plate for about 40 Ma. Also, the absence of rotation implies that the present-day strike of the NST of about 60°ENE is close to the original strike of the Late Cretaceous island arc in the Central Pacific (Fig. 5).

Finally, other paleomagnetic data from Northeast Asia on island arc complexes [6,7,27] infer the original positions of these complexes at northern mid-paleolatitudes of the Central Pacific [28,29]. Therefore, the existence of subduction zones and related island arcs and their subsequent large displacements are supported by other paleomagnetic data. Of course, many more results are still needed to better define the position of the mid-Pacific island arc system and the plate kinematics in this region.

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